

P1.0 Tunisia Purchase and Calibration Guide

Mostly printed sim-racing handbrake - Monday coupon plan, complete hardware list, local suppliers, and current price evidence

Revision 1 - 27 September 2026

Decision: print all eight coupon STLs once. They test different interfaces, not eight alternatives for one dimension. After the winning fits are measured, update the calibration parameters and regenerate P1.1 before printing the structural set.

The functional P1 prototype has **19 printed parts**. The number 18 belongs to the later P1-C commercial mixed-material design; do not use 18 when ordering or printing P1.0.

Budget snapshot

Scope	Planning cash requirement	Confidence
Mechanical hardware only	Approximately 95-170 TND	Engineering allowance; local per-piece quotes needed
Electronics and sensor	Approximately 45-60 TND	Most items have current local listing evidence
Complete purchased hardware	Approximately 150-230 TND	Excludes PETG, delivery, tools and failed prints
PETG consumed by final 19 parts	About 59 TND at 685 g / 86 TND per kg	Engineering mass estimate; actual slicer result controls [S21]
Recommended PETG stock	2 x 1 kg = about 172 TND	Includes coupons, development and reprint reserve [S21]
Startup cash with two PETG spools	Approximately 322-402 TND	Hardware plus development filament; excludes delivery/tools
Consumed value per successful unit	Likely below cash purchase	Packs, spare bearings and spring trials leave reusable stock

Prices and stock were checked online on 27 September 2026. A displayed price is not a reservation. Call before travelling, especially for exact bolt lengths, springs, magnets, and steel stock.

1. Monday: print and test all eight coupons

File	What it tests	Physical item required
COUPON_608_POCKETS_DEPTH_7.stl	608 fit at 7 mm bearing depth	Actual 608-2RS bearing
COUPON_608_POCKETS_DEPTH_14.stl	Deep/two-bearing pocket behaviour	Actual 608-2RS bearing
COUPON_M8_CLEARANCE.stl	8.0/8.2/8.4/8.6 mm pivot clearances	The actual M8 pivot bolt smooth shank
COUPON_M6_CLEARANCE.stl	6.0/6.2/6.4/6.6 mm bolt clearances	Representative M6 bolt
COUPON_M6_NUT_TRAPS.stl	10.0/10.2/10.4/10.6 mm AF nut traps	Actual M6 nuts
COUPON_MAGNET_POCKETS.stl	Nominal/+0.1/+0.2/+0.3 magnet fits	Final candidate magnet
COUPON_HALL_GAP_JIG.stl	3/5/8/10/12/15 mm sensing experiments	Magnet, SS49E, Pro Micro and multimeter
COUPON_LOADED_BEARING_WALL.stl	Representative bearing-wall strength and cracking	Bearing, M8 bolt, washers and hand load

After testing

- Record the selected bearing diameter, M8 and M6 hole, M6 nut-trap size, magnet pocket, and printer/material/profile.
- Measure the bearing OD/width, bolt shank diameter, nut across-flats, magnet dimensions, steel-bar thickness, and coupling-nut dimensions with calipers.
- Send the results back. P1 parameters must be updated and P1.1 regenerated. Do not print the old 19 structural files immediately after choosing a coupon.
- After regeneration, print the eight-part mechanical core first. Add sensor/electronics and the six clamp parts only after the core passes no-spring and light-spring tests.

2. Exact mechanical shopping list

Qty	Item/specification	Use and buying note	Price status
1	Mild-steel flat bar 25 x 5 x 280 mm	Main lever. Confirm actual 25/5 mm before P1.1.	QUOTE REQUIRED; allow 5-12 TND
2	Steel strips 25 x 3 x 180 mm	Base reinforcement; four 6.6 mm holes each.	QUOTE REQUIRED; allow 8-18 TND/pair
4	608-2RS, 8 x 22 x 7 mm	Two installed and two coupon/spares.	CONFIRMED 3.500 TND each; 14 TND total [S1]
1	M8 x 90 class 8.8 partially threaded bolt	Pivot. Smooth shank must cross both inner races.	EXACT LENGTH/SHANK QUOTE REQUIRED
1	8 ID x about 10 OD x 13.6 mm spacer tube	Between the two bearing inner races.	CUSTOM CUT / 3-8 TND allowance
2 sets	M8 washers/precision shims, 2.2 mm each side	Equal tower-to-bearing spacer stacks, max 16 mm OD.	MEASURE AND SELECT
1	M8 nyloc nut	Pivot retention; do not overtighten.	Local fastener shop
1 set	Extension springs, 15-20 mm OD, 60-100 mm hook length	Buy several light/medium rates. CAD hook distance is 69.76-85.82 mm.	QUOTE REQUIRED; allow 10-25 TND
1	M10 x 100 steel screw/rod	Desk clamp; steel thread only.	Local fastener shop; allow 3-7 TND
1	M10 coupling nut, about 30 mm, about 17 mm AF	Captured clamp thread. Measure before P1.1.	QUOTE REQUIRED; allow 3-8 TND
1	M10 swivel levelling foot/pad	Lower clamp contact, compatible with 46 mm carrier.	QUOTE REQUIRED; allow 5-15 TND
1+1	46 mm lower rubber and 90 x 62 x 2-3 mm upper rubber	Desk protection and grip.	ESTIMATED 5-10 TND

M6 structural fasteners

Qty	Size	Location
4	M6 x 35	Pivot towers through base/backing strips
2	M6 x 40	Steel lever through hub
2	M6 x 25	Spring module to base
2	M6 x 30	Fixed and moving steel spring anchors
1	M6 x 45	Moving travel-stop pin
2	M6 x 25 + rubber tips + locknuts	Adjustable release/full-pull stops
4	M6 x 30	Clamp interface through base
4	M6 x 30	Clamp butt-joint connections
2	M6 rig bolts, length to suit rig	Rig mounting with washers at least 18 mm OD
25+	M6 nyloc nuts	Buy spares
40+	M6 standard and large washers	Load spreading and adjustment

Budget 20-35 TND for the M6 assortment if bought locally by piece. Online benchmark: Brico Direct lists 10 M6 x 20 stainless bolts at 4.500 TND and 10 M6 nut/washer sets at 2.500 TND; the required lengths still need local confirmation [S2].

3. Small fasteners and electronics

Qty	Item	Use	Buying note
4	M4 x 35 + washers + nylocs	Two stop blocks	Buy 6
2	M4 x 30 + washers + nylocs	Hall fixed bracket	Buy 4
4	M4 x 20 + washers + nylocs	Electronics enclosure base	Buy 6
1	M4 x 40 + nyloc	Grip retention	Buy 2
4	M3 x 20 + nuts	Hall sled clamping	Buy 6
2 + 1	M3 x 12 and M3 x 20	Magnet holder/keeper	Buy spares
4	M3 x 10 + four M3 heat-set inserts	Electronics cover only	Heat-set inserts are not structural
2	M2.5/M3 board screws	Only after Hall PCB is measured	Do not pre-drill from an assumed board

Qty	Component	Current local evidence	Recommended action
1	Arduino Pro Micro 5 V / 16 MHz ATmega32U4	CoThings: 28.000 TND, listed in stock [S3]	Buy 1; USB cable not included
2	SS49E/49E linear Hall sensor	Didactico: 2.400 TND; 2B Trading: 2.450 TND [S4/S5]	Buy 2 so one is spare
1 pair	3-pin JST connector/cable	Didactico JST 2.54 pair: 1.500 TND [S6]	Verify pitch before soldering
1	Micro-USB data cable	Little Son 1 m data/charge cable: 7.500 TND [S7]	Must carry data, not charge-only
1	6 mm heat-shrink	Didactico: 2.200 TND [S8]	For 3-wire sensor loom
2	10 x 5 x 3 mm NdFeB block magnet	Exact Tunisian listing not confirmed	Current P1 pocket candidate; do not substitute silently
candidate	20 x 10 x 2 mm rectangular NdFeB magnet	Celectronix: 1.499 TND, listed in stock [S19]	Best locally listed rectangular candidate; requires revised magnet coupon/holder
candidate	12 x 4 mm circular NdFeB magnet	SELI: 1.800 TND, listed in stock [S20]	Compact local candidate; requires revised magnet coupon/holder
optional	10 x 2 mm N35 disc magnet	2B Trading: 1.200 TND [S9]	Experimental alternative only; requires matching pocket/calibration
as needed	Flexible 3-core wire, solder, cable ties	Didactico/CoThings/local electronics	Allow 5-10 TND

4. Where to buy in Tunisia

Supplier	Useful items	Address/contact	Status
Tunisie Roulements	608-2RS	6 Rue Chedly Kallala, Lafayette, Tunis 1002. 71 287 737 / 58 578 870.	608 listing and price confirmed [S1/S10]
Brico Direct	Fastener packs, nuts, washers, ties, tools	71bis Avenue Louis Braille, Tunis 1082. 71 100 950.	Prices are benchmarks; call for exact lengths [S2/S11]
SEGEMO - Maison du boulon	M3/M4/M6/M8/M10 bolts, nylocs, washers, shims	10 Rue Aid Jaber, Tunis 1000. 71 253 878.	Range confirmed; prices by counter quote [S12]
CoThings	Pro Micro and electronics	Route X, near Stade Bardo, Bardo, Tunis. 29 750 003 / 27 772 264.	Pro Micro price/stock listed [S3/S13]
Little Son	Pro Micro alternative, USB cable, M3 hardware	1 Rue de Piree, Tunis 1001. 58 114 788.	Pro Micro listed 35 TND; cable 7.5 TND [S7/S14]
Tuni Smart Innovation	SS49E, magnets, heat-shrink	1 Rue Pierre Mendes France, Ariana 2080. 51 954 443 / 51 954 448.	SS49E in stock; many magnets currently out of stock [S15]
Celectronix	20 x 10 x 2 mm rectangular magnet, electronics, PETG	Centre Said, Avenue Habib Bourguiba, Megrine. 79 295 570 / 27 582 469 / 28 581 332.	Magnet 1.499 TND; LUME PETG 1 kg 86 TND, listings in stock [S19/S21]
SELI	12 x 4 mm circular magnet and magnet alternatives	08 Rue Chedhly Kallala, 1st floor, Boumhel 2097. 29 002 608 / 92 168 725 / 55 560 037.	12 x 4 magnet listed in stock at 1.800 TND [S20]
Didactico	Hall, JST, heat-shrink, wire	Cite des Martyrs, Rue Mohamed Salah, Imm. Bouzguenda A01, Sfax. 54 776 776 / 99 707 685.	Listings/prices confirmed; delivery in Tunisia [S4/S6/S8/S16]
MTR Ressorts	Extension-spring prototype or small batch	ZI El Ons, Route de Tunis Km 10, Sakiet Ezzit 3021 Sfax. 98 333 883 / 98 331 896.	Local manufacturer; quotation required [S17]
ITI / Ressorts Mohamed Benaissa	Extension spring in Tunis	11 Rue Abderrahmen Azzem, Montplaisir, Tunis 1002. 71 908 311.	Local manufacturer; quotation required [S18]

What to say at the quincaillerie

Show the fastener table and say: 'Je veux des vis metriques zinguées ou classe 8.8, avec rondelles et ecrous Nylstop. Pour l'axe M8, il me faut une partie lisse qui traverse deux roulements 608; le filetage ne doit pas rouler dans les bagues.' Ask them to measure the smooth shank before you buy.

For the spring supplier: ask for a small selection of extension springs with closed hooks, 15-20 mm outside diameter and 60-100 mm free hook-to-hook length. Explain that the installed hook distance changes from about 69.8 to 85.8 mm and that the first unit is for force testing, not a production order.

For the magnet: the Celectronix 20 x 10 x 2 mm rectangular magnet is the closest useful locally listed shape, while SELI's 12 x 4 mm disc is the more compact option. Neither is a drop-in fit for the released 10 x 5 x 3 mm pocket. Buy one or two samples if convenient, measure them with calipers, and send the dimensions; only the magnet coupon and holder then need to be revised before structural printing.

5. Assembly purchase gates

Gate	Must be available/measured	Do not proceed if
Coupon gate	608, M8 sample, M6 bolt/nut, magnet, printer profile	You cannot physically select the fit
P1.1 regeneration	Winning coupon sizes and real hardware dimensions	The existing P1.0 fit assumptions are still unverified
Mechanical core	Base/towers/hub/grip, steel lever/strips, bearings, pivot stack, stops	Threads touch bearing races or towers spread
Spring gate	Light spring, steel anchors, positive mechanical stops	Spring becomes the travel stop or rubs printed parts
Clamp gate	M10 steel screw/nut, swivel foot, rubber pads	Any primary clamp thread is printed plastic
Sensor gate	Magnet mechanically retained, fixed SS49E, 3-wire loom	Magnet or sensor can collide with the lever
Electronics gate	Pro Micro and data-capable USB cable	Windows does not enumerate the USB HID device

Important limitations

- Do not buy a precision production spring before the first pull-force test.
- Do not allow threaded M8 sections to act as the bearing journal if a smooth-shank bolt or 8 mm shaft is available.
- Do not print all 19 parts from the uncalibrated P1.0 export merely because one coupon looks correct; all critical coupon results must be transferred into P1.1.
- The listed prices exclude delivery and may change. 'Confirmed' means the retailer displayed the item/price at research time, not that stock has been reserved.
- Structural capacity remains a physical-test question. Stop at cracking, whitening, permanent deformation, bearing movement, or fastener loosening.

6. Source register

S1 - Tunisie Roulements - 608-2RS, 3.500 TND

<https://tunisie-roulements.tn/produit/roulement-608-2rs/>

S2 - Brico Direct - visserie listings and pack prices

<https://brico-direct.tn/208-vis-et-boulons>

S3 - CoThings - Pro Micro 5V/16MHz, 28.000 TND

<https://cothings.net/products/arduino-pro-micro-mini-usb-atmega32u4-5v-16mhz>

S4 - Didactico - SS49E, 2.400 TND

<https://didactico.tn/categorie-produit/capteurs-prototypage/capteurs/capteurs-magnetiques-hall/>

S5 - 2B Trading - SS49E, 2.450 TND

<https://2btrading.tn/transistor-igbt/4950-capteur-effet-hall-49e-oh49e-ss49e.html>

S6 - Didactico - JST 3-pin pair, 1.500 TND

<https://didactico.tn/categorie-produit/connectique-cables-boitiers/>

S7 - Little Son - 1 m Micro-USB data cable, 7.500 TND

<https://little-son.tn/2-accueil?page=2>

S8 - Didactico - 6 mm heat-shrink, 2.200 TND

<https://didactico.tn/produit/gaine-thermoretractable-drs-6/>

S9 - 2B Trading - 10 x 2 mm N35 disc magnet, 1.200 TND

<https://2btrading.tn/2-accueil?page=80>

S10 - Tunisie Roulements contact

<https://tunisie-roulements.tn/contact/>

S11 - Brico Direct store/contact

<https://brico-direct.tn/magasins>

S12 - SEGEMO contact

<https://www.segemo.tn/contact/>

S13 - CoThings store location

<https://cothings.net/pages/emplacement-du-magazin>

S14 - Little Son Pro Micro listing, 35.000 TND

<https://little-son.tn/accueil/97-arduino-pro-micro-atmega-32u4.html>

S15 - Tuni Smart SS49E and store address

<https://tuni-smart-innovation.com/products/capteur-a-effet-hall-lineaire-ss49e-oh49e>

S16 - Didactico store/contact

<https://didactico.tn/a-propos-didactico/>

S17 - MTR Ressorts

<https://mtr-ressorts.tn/fr>

S18 - ITI Tunisie

<https://iti.com.tn/presentation/>

S19 - Celectronix - 20 x 10 x 2 mm rectangular neodymium magnet, 1.499 TND, listed in stock

<https://www.celectronix.com/autres-mecanique/7498-aimant-cylindrique-de-n-odyme-16-mm-de-diam-tre-x-1-8-mm-d-paisseur-copie-.html>

S20 - SELI - 12 x 4 mm circular neodymium magnet, 1.800 TND, listed in stock

<https://seli.tn/product-category/aimant-et-electro-aimant/>

S21 - Celectronix - LUME PETG 1.75 mm, 1 kg, 86.000 TND, multiple colors listed in stock

<https://www.celectronix.com/filament-petg/8156--074-0-barre-led-tv-lg-43uk6565-7led-3v-44-5cm.html>

Prepared from the released P1.0 BOM and current public Tunisian retailer/manufacturer pages. This is a prototype purchasing plan, not a supplier quotation or guarantee of compatibility.